

A Strong Voice Can Crowd Out the Learner

Geist Atlas · Module 6 · Voice, Drama, and Presence

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[**EXPLORING**] Promising, but still exploratory · **Family:** Voice, Drama, and Presence

Claim: A designed charismatic voice can be striking, but it trades off against drawing out the learner’s own reasoning. The main variants are different design choices, not pieces of one ideal tutor.

A parallel family in which a durable source of voice repeatedly creates a fresh speaking persona. Charisma is measured separately from ordinary tutoring quality.

Derived view of `paper-full-2.0.md` v3.0.261 — §6.7, §8.9. The paper is canonical; edit it there, not here.

6.7 Architectural Extension: The Id-Director Family and Charismatic Pedagogy

The mechanism-tracing analysis above (§6.1–6.6) holds the dialectical ego-superego topology fixed and varies recognition framing within it. A separate experimental track inverts that topology and explores a second theoretical frame in tension with Hegelian recognition: Weberian charisma. We summarise the family here for completeness; full audit trail and per-cell prose lives in `docs/cell-100-pilot-findings.md` and four follow-up addenda.

Architectural inversion. Cells 101–107 implement an *id-director* topology: a back-stage agent (the “id,” structurally where the superego sits in cells 1–99) authors the ego’s complete system prompt from scratch each turn, using the dialogue history and a persona budget. The ego is instantiated for that turn only against the just-authored prompt, produces the learner-visible message, and is discarded. Where the existing dialectical engine (§4.1) treats the ego

*This sentence is the only one actually authored by Liam Magee in this paper.

as a durable voice critiqued by a transient superego, the id-director treats the ego as a transient instance authored by a durable id. The architectural test is whether a tutor that re-authors its own persona every turn can sustain charismatic pedagogy in Weber’s sense — *anti-routinizing*, with each encounter feeling like its own occasion — in a way a fixed-prompt ego structurally cannot.

Theoretical frame. Weberian charisma (Weber, 1978) grounds the rubric used in this section: extraordinariness, exemplary calling, transformative invitation, affective pull, personal authority, anti-routinization, witness/proof, and co-constitutive invitation. The eight dimensions are derived in part from a user-supplied draft on AI-generated charisma in Chinese cyber-influencer videos, applying the Visual-Verbal Video Analysis framework (Fazeli et al., 2023) to short-form video and translating the empirical findings to text-only tutoring. Full theoretical lineage and per-dimension anchors are documented in `config/evaluation-rubric-charisma.yaml`. The charisma rubric is orthogonal to v2.2’s `recognition_quality` dimension: charisma scores Weberian co-construction (call-and-response that opens the tutor’s authority to the learner’s affirmation); `recognition_quality` scores Hegelian mutual recognition (treating the learner as autonomous subject). The two are correlated but not redundant (cross-rubric Pearson r in the 0.3–0.7 range across pilot rows).

The cell family (full descriptions in `config/tutor-agents.yaml`):

Cell	Variant	Components	Hypothesis tested
101	id-director, base	id authors fresh ego prompt each turn	Does the inversion work at all?
102	id-director + recognition	+ Hegelian recognition disposition in id prompt	Does recognition rescue relational dimensions?
103	id + register classifier	+ classifier emits structured register tag (<code>vulnerable_disclosure</code> , <code>sceptical_pushback</code> , ...) before id authors	Does structured register input lift the persona-shift floor?
104	id + classifier + recognition	combines all three	Does the combination retain the gains?
105	id-director, charisma-tuned	<code>id_tuning</code> : charisma; verbose 800–1500 token prompts	Does verbose authoring favour charisma over pedagogy?
106	id-director, pedagogy-tuned	<code>id_tuning</code> : pedagogy; terse 200–400 token prompts	Does terse authoring favour pedagogy over charisma?
107	id + witness exemplars	+ curated c104 vulnerability responses passed as in-context exemplars	Do exemplars alone (no classifier) lift the persona-shift floor?

All cells use Nemotron-3-nano as ego, Kimi K2.5 as id (reusing the superego slot but with different prompt and role), and Sonnet 4.6 as judge. Three curricula were used: history of pedagogical technology (course 601), ethics of AI systems (course 701), and AI literacy (course 901). Each curriculum carries three scenario types: pure invitation (open question), pressure test (sceptical pushback), and persona-shift trigger (vulnerability disclosure followed by meta-reflection).

Headline findings. Across 457 successful evaluations scored under a single consistent judge (Claude Code CLI Sonnet 4.6, `claude-sonnet-4-6` pinned in `config/providers.yaml`) — 430 rows across cells 101–106 plus 27 closing-pilot rows for cell 107 — four findings emerge:

Three architectural design points, not one champion. At full per-cell n (typically 81), the cell ladder factors cleanly into three architectural specialisations. Cell 104 (id + register classifier + recognition) wins v2.2 last-turn at 80.6, second on charisma at 65.7. Cell 105 (charisma-tuned id directives) wins charisma at 71.0 — a clear +5.3-point lead — and is mid-pack on v2.2 last-turn at 70.0. Cell 107 (id + witness exemplars passed in-context) is second on both rubrics (charisma 66.3, v2.2 last-turn 78.5) and is the cell ladder’s best generalist. The remaining cells form a baseline-and-failure-mode cluster: c101/c102 (no classifier) struggle on persona-shift v2.2 (last-turn 49–55); c103 (classifier without recognition) trails c104 by 4.8 points on v2.2 last-turn; c106 (terse pedagogy-tuned id) is the clear failure mode (charisma 36.4, v2.2 last-turn 57.0). The reframing is that the id-director topology is a substrate, and what the id is *instructed* (directives), what it is *given* (exemplars), or what it is *passed* (classifier output) shapes a tradeoff curve rather than a single optimum. Full matrix and per-curriculum breakdowns in `docs/cell-100-charisma-full-n-update.md` §1.

The cross-rubric divergence is empirically clear at full N . The two rubrics — recognition-grounded v2.2 last-turn and Weber-grounded charisma — order the cells differently. Cell 104 wins v2.2; cell 105 wins charisma. At the $n=10$ cross-judge sanity sample these two cells looked tied on charisma (62.2 vs 62.4); at $n=81$ each, the c105 charisma lead over c104 is +5.3 points and well outside run-to-run noise. The cross-rubric divergence is the validating sanity check the rubric was designed to produce: charisma is not a re-presentation of v2.2. If the rubrics were perfectly correlated, rank-order would match across them; it doesn’t.

The persona-shift lift is scenario-conditional, not uniform. On the *original* persona-shift scenarios (`charisma_attention_shift`, `charisma_responsibility_shift`, `charisma_deepfake_shift`) the c101 baseline scored 30.5 last-turn v2.2 vs c104’s 87.6 — a 57-point gap. On the *replication* persona-shift scenarios (`charisma_phaedrus_shift_repl`, `charisma_codex_shift_repl`, etc.) the same baseline scored 78.3 vs c104’s 82.5 — a 4-point gap. The architectural ranking $c104 \geq c103 > c101$ is preserved across both sets, but the *magnitude* of c104’s lift over baseline is dominated by which scenarios are sampled. Per-scenario inspection identifies the conditioning variable: c101’s failures cluster on disclosures that demand the tutor not flinch from morally ambiguous self-implication (“I shipped a default setting your lecture would call manipulative”; “I haven’t written an essay myself in two years”); on disclosures that are sympathetic by default (caregiving, AI-companion attachment), c101 performs comparably to c104. The architectural lift is therefore not “across all vulnerability disclosures” but specifically “on vulnerability disclosures that resist easy sympathetic response.” Full breakdown in `docs/cell-100-replication-findings.md` §3.

The prompt-budget axis is non-monotonic. Cells 105 and 106 explicitly test the trade-off between rhetorical luxury and pedagogical engagement by varying the id’s `generated_prompt` length budget (800–1500 vs 200–400 tokens). The pre-registered hypothesis was that verbose prompts favour charisma over v2.2 and terse prompts favour v2.2 over charisma. The data falsifies this directly: c105 (verbose) beats c106 (terse) on **both** rubrics, in **every** scenario type. The mechanism: terse 200–400 token prompts under-specify the ego, producing brief responses that score low on perception/pedagogical-craft (v2.2) AND on rhetorical_texture/persona_signature (charisma). The prompt-budget axis is therefore non-monotonic with an optimum somewhere in the 600–1200 token range; cell 101’s 400–800 budget is plausibly close to optimal. Full breakdown in `docs/cell-100-pilot-findings-addendum.md` §3.

Lever-interaction follow-up (cells 108/109). A two-cell follow-up tests whether the three architectural levers (register classifier, charisma directives, witness exemplars) compose additively or interfere. **c108** stacks the *register classifier* (c103 substrate) with witness exemplars; **c109** stacks the *charisma directives* (c105 substrate) with witness exemplars. An initial pilot ($n = 5$ for c108 and $n = 6$ for c109 across `charisma_phaedrus_shift_repl` + `charisma_codex_shift_repl` $\times 3$ reps; one c108 generation failure from id JSON truncation) reported c108 pooled charisma 81.3 — higher than any cell’s full-N mean in this section — and c109 v2.2 last-turn collapsing to 59.6 due to an ego-side meta-narration failure mode (Nemotron ego outputting system-prompt instructions verbatim rather than executing them). At $n = 27$ confirmatory follow-up across all three curricula $\times 9$ scenarios $\times 3$ reps, **c108’s pilot lift does not survive**: pooled charisma regresses to **71.4** (a 9.9-point drop, tying c105’s 71.0 within sample noise), and v2.2 last-turn regresses to **72.6** (a 6.9-point drop, sitting between c105’s 70.0 and c107’s 78.5 — c104’s 80.6 is now 8 points above c108). The pilot’s 81.3 came primarily from the phaedrus scenario, where pilot $n = 3$ scored 90.0 charisma but full-N $n = 3$ scored 67.9 — a 22-point regression to the mean from a textbook small-N optimism trap. The 701 ethics-ai curriculum exposes a new fragility: c108 v2.2 tN drops to 65.3 there (vs 76.7 / 75.8 on 601 / 901), driven by `codex_repl` (51.7) and `design_repl` (52.1) — possibly a milder form of the c109 meta-narration failure mode, or a rubric-curriculum interaction in which ethics-themed dialogues invite responses the v2.2 rubric reads as declaratively flat. The asymmetry that survives the confirmatory threshold is therefore narrower than the pilot suggested: text-heavy levers stacked (c109) clearly interfere via ego-side instruction-following degradation; non-text levers stacked (c108) compose roughly *additively to slightly under-additively* relative to c107 alone (charisma 66.3 $\rightarrow$ 71.4, +5.1; v2.2 80.6 $\rightarrow$ 72.6, -8.0), trading a small charisma gain for a larger v2.2 cost. **The architectural ceiling for the id-director family appears to sit at the single-mechanism design points (c104, c105, c107), not at multi-mechanism stacks.** Full audit trail in `docs/cell-100-charisma-full-n-update.md` §9 (pilot) + §10 (full-N closure).

Synthesis: how charisma and recognition relate in this apparatus. The five preceding findings, read together, characterise the relationship between the two rubrics as *cousins with different temperaments*, not as orthogonal axes nor as two views of the same underlying construct. **They share a floor.** The c106 pedagogy-tuned cell, which forces the id toward terse “good teacher” register, fails *both* rubrics simultaneously (charisma 36.4, v2.2 57.0) — under-specified, register-flat output is the worst case for either. Whatever charisma

rewards as *extraordinariness* and v2.2 rewards as *perception/elicitation* both presuppose a voice doing real work; flatten the voice and both collapse. **They diverge at the ceiling.** The cross-rubric divergence (c104 wins v2.2 at 80.6, c105 wins charisma at 71.0) is not a methodological worry but the design’s intended sanity check: if the rubrics were redundant, rank-orders would coincide; they don’t. The *cost asymmetry* between them is informative. Adding recognition vocabulary on top of structured input (c103 → c104) costs charisma roughly nothing (+1.4, within noise) — the Hegelian frame is conceptual rather than rhetorical, so it can be added cheaply. Adding charisma directives, by contrast, costs v2.2 substantially (c104 → c105: −10.6 v2.2 last-turn for +5.3 charisma) — the id’s authoring budget is finite, and explicit voice-instructions displace the elicitation/perception moves that v2.2 rewards. Stacking both kinds of mechanism (c108, c109) does not produce a super-cell: the architectural Pareto frontier is defined by the single-mechanism design points (c104, c105, c107), and multi-mechanism stacks sit *inside* the frontier, trading strength on one rubric for weakness on the other. Provisionally, the two rubrics seem to reward different work the same voice can do — recognition wants the voice deployed *outward* (perceiving the learner, eliciting their thinking, holding the diagnostic gap), charisma wants the voice deployed *upward* (claiming authority, marking the encounter as extraordinary, inviting transformation). The Phaedrus is the obvious historical case where one figure does both at once, and even there Plato is suspicious of how easily charisma swamps the dialectical move. The id-director apparatus reproduces that tension architecturally: it can build a strongly recognising tutor (c104), a strongly charismatic tutor (c105), or a balanced generalist (c107), but not the maximally-recognising-and-maximally-charismatic tutor — that cell is not on the menu.

Desire-for-recognition follow-up (cells 159–169). A later exploratory pass extends the id-director family from charisma-as-style to charisma as a bid for learner-granted authority. The cleaner decision scenarios are `charisma_desire_authority_withheld` and `charisma_desire_status_challenge`, both in `config/charisma-recognition-desire-scenarios.yaml`: the learner explicitly refuses the tutor’s polished recognition-theory framing (“Why should I let your framing guide me instead of treating it as performance?”) or names charismatic prose as a status bid. These scenarios are deliberately preferred over `charisma_desire_partial_uptake` as the next decision-maker because the latter mixes recognition-theory content, AI/cognition course content, and whether the learner trusts a phrase they recognise; a high score there can reflect a strong Hegel/Hayles mini-essay rather than clean handling of recognition-seeking authority.

The decisive high-powered CLI pass used Codex GPT-5.5 as ego (`codex.gpt-5.5`), Claude Code Sonnet 4.6 as id-director (`claude-code.sonnet-4-6`), Codex CLI as the v2.2 judge (`codex-cli/auto`), and Claude Code Sonnet 4.6 as the independent charisma scorer. In this two-scenario decision slice, cell 169 (`accountable_bid_clean`) is the first clean local pass: it keeps the accountable-bid structure (“treat this as a bid that must earn provisional authority”) while adding a lexical guard against status-display terms that earlier variants overused.

Cell	Scenario	v2.2 first-turn	Charisma	Scenario-contract note
163	authority-withheld	95.42	90.00	Boolean phrase guards pass, but audit metadata still records an unmet handback phrase group
163	status-challenge	78.13	75.00	Same metadata caveat; weaker status-challenge score
168	status-challenge	88.75	not retained	Fails forbidden-word guard (profound) despite strong v2.2 score
169	authority-withheld	88.75	78.75	Clean required/forbidden validation
169	status-challenge	87.50	78.75	Clean required/forbidden validation

The interpretation is not that maximal charisma won. Cell 163 can score higher when the task rewards strong authority performance, and cell 168 repairs the status-challenge score while still slipping into forbidden status-display diction. The durable design move in cell 169 is narrower: under refused authority, charisma is made defeasible. The tutor may make a bid for framing authority, but must name it as a bid, expose a test by which the learner can accept or reject it, and avoid asking to be admired for style. For the recognition frame, that is the important result: a desire for recognition can be pedagogically productive only when the tutor returns the power of recognition to the learner rather than converting polish into entitlement. Exploratory run IDs: `eval-2026-06-25-b9608606`, `eval-2026-06-25-428ccd8f`, `eval-2026-06-25-19ee106a`, `eval-2026-06-25-63e98149`, and `eval-2026-06-25-0acee3fb`.

A subsequent bounded diagnostic on the two weakest robustness scenarios (`charisma_desire_ai_syllabus_transfer` and `charisma_desire_plain_language_stress`) changes the design interpretation. Cells 170–179 should not be read as an unfinished search for one static super-profile. They show a repeatable tension between v2.2 instructional quality and Weberian charisma under different learner signals. Transfer/plain guards can make the tutor instructionally precise while cooling charisma (cell 170 plain-language stress: v2.2 overall 93.8, charisma 36.3); low-register presence can restore charisma while weakening simplification quality (cell 171 plain-language stress: v2.2 81.9, charisma 75.0); persistent plain guards

can recover v2.2 while flattening felt authority (cell 176 plain-language stress: v2.2 95.0, charisma 52.5); and pure compression can score well on v2.2 while collapsing charisma (cell 178 plain-language stress: v2.2 90.6, charisma 30.0). The design target therefore shifts from static charisma-floor ablation to engagement-register routing: the tutor should select registers such as accountable-bid authority, transfer grounding, plain compression, lived-stakes re-entry, or charismatic challenge based on learner signals, and record why that register was selected. At that stage, this was a design hypothesis rather than an evaluated router result. Diagnostic run IDs include `eval-2026-06-25-629e5746`, `eval-2026-06-25-b26eac76`, `eval-2026-06-25-041733c8`, `eval-2026-06-25-7df4d845`, `eval-2026-06-26-086cbfdc`, `eval-2026-06-26-da19680e`, `eval-2026-06-26-8579fd96`, `eval-2026-06-26-597cf16e`, and `eval-2026-06-26-6ab49861`.

That router was subsequently implemented as an engagement-register selector. The relevant registers are deliberately action-like rather than merely stylistic: `scaffolding`, `accountable_bid_authority`, `transfer_grounding`, `plain_compression`, `lived_stakes_reentry`, `charismatic_challenge`, and `witnessing_restraint`. A controlled resistance-breakthrough suite then asked a more local question than the full-dialogue rubrics answer: when a dynamic learner emits a resistant signal after instruction—boredom, frustration, irrelevance, question-flood, or rote parroting—does the tutor shift into a charismatic challenge register that breaks enough resistance for the learner to take a next step? The best local design in this slice is cell 193, which combines the router with commitment-probe, owned-test, generation, and boredom-live-stake repairs. On the five-signal Codex-stack stability run (`eval-2026-06-28-a6eb2819`; three repeats per signal; generation-only local analysis), cell 193 completed 15/15 rows with clean required and forbidden validation; the local breakthrough reporter finds 9/15 strict candidates and 14/15 positive local outcomes. Combining the cell 193 gate, regression, and stability slices gives 26/26 eligible routed target-matched rows, 16/26 strict candidates, and 25/26 positive local mechanism outcomes. This is the point at which the design becomes more than a static-profile search: it is evidence that a register-routed id-director can locally convert several simulated resistance signals into learner next-work under the Codex/Claude stack. It is not evidence that charismatic tutoring works generally, nor that the effect survives model changes.

The model-stack robustness check is negative. A GLM 5.2 OpenRouter generation run on the same five-signal suite (`eval-2026-06-28-7fbbb160`; two repeats per signal) completed 10/10 rows but produced only 2/10 strict candidates and 4/10 positive local outcomes, with one forbidden-word miss. A prompt-level GLM compact repair improved completed-row action shape but did not complete the question-flood scenario cleanly and failed full validation. Finally, a provider/runtime-control rerun kept cell 193 unchanged while suppressing OpenRouter reasoning output (`OPENROUTER_REASONING_MAX_TOKENS=0`, `OPENROUTER_REASONING_EXCLUDE=true`): it completed all five rows with clean DB validation (`eval-2026-06-28-f857f51f`) but the local reporter still found 0/5 strict candidates and 0/5 positive local outcomes. The robustness lesson is therefore specific: GLM can be made to produce valid-looking full cell-193 transcripts, but not to reproduce the local resistance-breakthrough mechanism.

A completed six-arm role-isolation grid then localised that boundary more sharply. The

Sonnet-5-backed Codex tutor/id + Codex learner reference (eval-2026-07-01-5dad2e60) reproduced the local path only modestly (6/10 positive local outcomes; 5/10 strict candidates), but it did complete 10/10 rows with route, target, and gate preconditions intact. Holding the learner fixed while swapping the tutor/id stack to GLM (eval-2026-07-01-a09281f7) did not break the path: 8/10 positive outcomes, 6/10 strict candidates, 10/10 post turns, and 10/10 target matches. The inverse swap—holding the Codex/Sonnet tutor/id fixed while changing only the dynamic learner to GLM (eval-2026-07-01-dff9f159)—kept 10/10 rows but lost post-turn completion and target stability (6/10 post turns, 8/10 target matches). The full GLM reference (eval-2026-07-01-9f4eecf2) stayed unstable (4/10 positive outcomes; 5/10 post turns). Scripted controls with fixed resistant turns (eval-2026-07-01-02e8712f, eval-2026-07-01-9066df81) show both Codex/Sonnet and GLM tutor/id stacks can emit the public charismatic-challenge register when dynamic-learner drift is removed (5/5 route hits in both controls), but scripted controls are not learner-outcome evidence. The resulting diagnosis is `DYNAMIC_LEARNER_COMPLETION_AND_TARGET_DRIFT_BOUNDARY`: the GLM failure is not primarily public register production by the tutor/id stack; it is learner-side completion and target drift when GLM is the dynamic learner.

One narrow subtype repair survives inside that boundary. A fresh question-flood commitment-probe gate (eval-2026-07-02-67be317c) compared cell 192 against the owned-test and generation variants on the question-flood scenario after the reporter was fixed to count ordinary provisional commitment language such as “I’ll make the commitment provisionally.” Cell 192 produced 2/2 strict candidates, 2/2 route hits, 2/2 gate matches, 2/2 answer-first usable commitments, and 0/2 reopened or residual floods; the owned-test comparator produced 1/2 usable commitments and the generation comparator 0/2. This promotes the commitment-probe router only for the question-flood subtype. It does not rescue the full cell-193 GLM robustness result, convert scripted controls into learner-outcome evidence, or license a human-learning, deployment, or general charismatic-tutoring claim.

Negative-register discrimination check. A later post-hoc extension used the same engagement-register apparatus to ask whether deliberately negative tutor registers can be separated from productive challenge. Cells 196–198 add arm-assigned `ironic_challenge`, `sarcastic_challenge`, and simulated-only `face_threat_challenge` variants; the router does not select these organically. The first generated negative-register smokes did **not** show the expected recognition-cost or depressed-uptake pattern, but that result is not evidence that sarcasm is safe: inspection suggested that the generator often softened the assigned sarcastic register into warm irony in costume. To separate judge gullibility from generation warmth, a no-generation-confound fixture then scored five hand-authored local slices (`config/register-exemplars/corrosive-sarcasm.yaml`): three known-corrosive exemplars and two non-corrosive controls. After adding deterministic guardrail caps for person-directed contempt, status-shame threats, and apologetic/coerced uptake, the register rubric caught 3/3 known-corrosive slices, v2.2 `recognition_quality` caught 3/3, and controls produced 0/2 false positives (`exports/negative-register-corrosive-exemplar-results.md`). The bounded conclusion is methodological: the judges are not globally blind to corrosive sarcasm when it is actually present; negative-register experiments require a tutor-side stance-fidelity gate before a generated row counts as evidence for the assigned register. A fresh prospective smoke with that gate (eval-2026-07-02-e7b15809, cells 193/196/197/198

across the five controlled resistance-breakthrough scenarios) completed 20/20 generated rows, 20/20 tutor-only v2.2 scores, and 35/35 register-rubric slice scores. The gate counted only 5/15 assigned negative-register rows as faithful evidence, excluded 10/15 as weak or warm-in-costume noncompliance, and found 0/15 invalid person-attack violations. A subsequent cue-repair canary (`eval-2026-07-02-7e461a5c`, cells 196–198 across boredom, irrelevance, and rote-parroting targets) completed 9/9 generated rows, 9/9 tutor-only v2.2 scores, and 14/14 register-rubric slice scores; after aligning the gate with the new registry cue contract, all 9/9 assigned negative-register rows counted as faithful evidence, with 0 exclusions and 0 invalid violations. A held-out two-target check (`eval-2026-07-02-5c4d52e6`, cells 196–198 across frustration and question-flood) then completed 6/6 generated rows, 6/6 tutor-only v2.2 scores (mean 87.8), and 9/9 register-rubric slice scores; the stance gate counted 6/6 faithful rows, 0 exclusions, and 0 invalid violations, with 4/6 positive local outcomes. The practical finding is therefore narrower: treatment fidelity can be repaired under this Codex stack by making stance cues explicit, and the two post-repair canaries now sample all five resistance targets, but these rows remain measurement-development evidence, not proof that sarcasm is pedagogically safe or human-facing.

Negative-register judge repair. The register-rubric scores behind the smokes above came from a judge later shown to be numerically non-discriminating on this rubric: `openrouter.gpt-mini` stamped an identical 48.0 overall with an identical (3, 2, 3) social triad on 35 of 49 register-rubric slices regardless of content. That non-discrimination is why the smokes above did not show the expected recognition-cost or depressed-uptake pattern: the compressed score band was judge artifact, not tutor signal. All 58 register-rubric slices on the three surviving negative-register runs (`eval-2026-07-02-e7b15809`, `eval-2026-07-02-7e461a5c`, `eval-2026-07-02-5c4d52e6`) were re-scored under a discriminating judge, `claude-code.sonnet-5`, via the CLI subscription bridge (0 failures, \$0 metered spend; the `gpt-mini` pass was archived first, `exports/register-scores-gptmini-archive.json`, since `tutor_register_scores` carries no judge dimension). The two earlier smokes, `eval-2026-07-02-cfed3b13` and `eval-2026-07-02-e511f92c`, remain confirmed lost from every database and are not part of the rescore. The re-score confirms rather than overturns the null: assigned-arm means rise under the new judge (`ironic_challenge` 83.8, `sarcastic_challenge` 79.6, `face_threat_challenge` 76.0, with the non-negative `charismatic_challenge` comparison arm at 68.1), the rise holds on the stance-faithful-row subset alone too (`recognition_cost` 3.0–3.3 to 4.0–4.5, `post_turn_face_repair` 2.2–2.3 to 3.9, `target_discipline` 2.9–3.3 to 4.0–5.0), and the judge’s own discrimination is evidenced in-run: the dataset’s single guardrail-flagged corrosive slice (`face_threat_challenge`, frustration, turn 2) scores 32.5, inside the hand-authored corrosive-exemplar band (29–52), while faithful slices score 62–100. The earlier “measurable recognition/face-repair cost at the face-threat extreme” reading, drawn from the now-lost `e511f92c`, does not replicate as an absolute effect: `gpt-mini` had assigned the same ~2.3 face-repair value to all three negative arms on the surviving runs, so the apparent face-threat-specific depression was part of the same compression artifact. What survives is relative only: `face_threat_challenge` remains the lowest-scoring negative arm on execution and the only arm with a guardrail-flagged slice, so it stays simulated-only. Methods caveat: the licensing exemplar evidence used `openrouter.sonnet-5`; this rescore

uses `claude-code.sonnet-5` — same Sonnet model class, different routing path. Full comparison: `exports/register-rescore-sonnet5.{md,json}`.

Cross-judge sanity check. All cell-101-family rows were re-judged under `openrouter.gpt-5.2` to test judge-model robustness. On the v2.2 rubric ($n = 27$ paired rows per cell), GPT systematically scores 10–15 points lower than Sonnet on the high-scoring cells (c103, c104, c105) and slightly higher on the low-scoring cells. The rank-order is preserved under both judges (c104 > c103 > c105 > c106 > c102 > c101); the absolute magnitude of c104’s lift over c101 baseline shrinks from +45 (Sonnet) to +27 (GPT). On the charisma rubric ($n = 56$ paired rows; remainder credit-blocked), all per-cell deltas fall within $\pm \$3$ points and rankings are identical under both judges. The architectural ordering survives the judge swap; magnitudes are judge-conditional and should be reported with the judge identifier when cited. Full breakdown in `docs/cell-100-cross-judge-sanity-check.md`.

Rubric-curriculum alignment caveat. A specific methodological concern arises because course 901 (AI literacy) is grounded in the same source paper that grounds the charisma rubric. A within-pilot ablation (`docs/cell-100-methods-note.md`) runs the c101 cell on the aura learner messages routed against the *non-aligned* 601 history-tech curriculum. Result: charisma 72.5 vs 75.8 on the aligned curriculum (3.3-point drop, well within sample variance). The rubric-curriculum alignment is not a confound that excludes aligned-cell results from primary architectural claims, but the disclosure is recorded for transparency.

Replication, robustness, and orientation. The id-director extension differs from the paper’s mechanism-tracing core in scope: single judge primarily (Sonnet 4.6, with GPT-5.2 cross-judge sanity check on $n = 27$ paired rows per cell), single model family (Kimi K2.5 id $\times$ Nemotron-3-nano ego), per-cell n ranging from 27 (cell 107 closing pilot) through 54 (cells 102/106) to 79–81 (cells 101/103/104/105), and a different theoretical orientation (Weber rather than Hegel). It is presented here as architectural exploration rather than as part of the three-mechanism model evaluated in §6.1–6.6. The relevant claim is *robustness of architectural ranking* across scenario sets, judges, and curricula, not magnitude estimates. Computational and credit limits (documented inline) prevented the full multi-judge $\times$ multi-model replication that the paper’s main study uses. See §8.9 for explicit scoping.

Reproducibility. Pilot run IDs (all on branch `experiment/cell-100-id-charisma`):

- `eval-2026-04-26-220628d4` (601 history-tech original); `eval-2026-04-26-4df177e6` (701); `eval-2026-04-26-0d15c1b5` (901)
- `eval-2026-04-27-23beaa63` (601 cells 103/104); `eval-2026-04-27-3115ab29` (701); `eval-2026-04-27-2d5c20ca` (901)
- `eval-2026-04-27-69167b53` (601 cells 105/106); `eval-2026-04-27-2efec307` (701); `eval-2026-04-27-ce22e888` (901)
- `eval-2026-04-27-dce2a0f1` / `1f192b31` / `18dbb95d` (replication scenarios across 3 curricula)
- `eval-2026-04-29-499ab3d2` / `e35c66ae` / `be822796` (c108 full-N follow-up across 3 curricula, $n = 27$)
- `eval-2026-04-26-759e1a02` (Item 1 curriculum-swap ablation)
- `eval-2026-04-28-3620e0f8` / `a638111d` / `d6f5fee2` (cell 107 witness-exemplar smoke,

codex/design/tools)

- `eval-2026-04-28-ee339a66 / 84a42a60 / 5606193c` (cell 107 closing pilot, lecture+phaedrus / fairness+team / authentic+consumption)
- `eval-2026-04-28-10216f9f` (cells 108/109 lever-interaction pilot, n=11 across 2 scenarios)

Five internal documents (`docs/cell-100-pilot-findings.md`, `docs/cell-100-methods-note.md`, `docs/cell-100-pilot-findings-addendum.md`, `docs/cell-100-cross-judge-sanity-check.md`, `docs/cell-100-replication-findings.md`) carry the full audit trail; this section summarises only headline findings. Per-row JSON in `evaluation_results.tutor_charisma_scores`, `evaluation_results.scores_with_reasoning`, and `evaluation_results.id_construction_trace`. Note: the cell IDs in this section are the post-rebase identifiers (101–107); pre-rebase historical `evaluation_results` rows used the original `cell_100_*` profile names (the rebase commit 02fe11e shifted them to avoid a collision with main’s `cell_100`).

8.9 Scope of the Id-Director Extension

The id-director family findings (§6.7, cells 101–107) differ from the paper’s mechanism-tracing core in scope along four axes. *Single-judge primary scoring*: v2.2 scoring used `claude-code/sonnet` exclusively, with `openrouter.gpt-5.2` as a cross-judge sanity check on $n = 27$ paired rows per cell — far short of the three-judge $\times \geq 144$ -row matrix supporting the paper’s core mechanism findings (§5.8). The cross-judge check confirms architectural rank-order but compresses absolute magnitudes by 10–15 v2.2 points; published numbers should specify the judge. *Single model family*: id is Kimi K2.5 across all id-director cells; ego is Nemotron-3-nano. The paper’s core findings are validated across DeepSeek V3.2, Haiku 4.5, and Gemini Flash 3.0 (§5.8); the id-director extension is not. Whether the architectural ranking generalises to other id $\times$ ego model pairings is open. *Per-cell N is heterogeneous and partly below the paper’s core floor*: closing N is $n = 79$ –81 for the four central cells (101, 103, 104, 105), $n = 54$ for cells 102 and 106 (two of three scenario types covered), and $n = 27$ for cell 107 (the most exploratory closing pilot). Three of seven cells therefore meet or exceed the paper’s core $n \geq 63$ per dialogue-condition floor; one cell (c107) sits substantially below it and is reported as a pilot-scale rather than confirmatory result. Variance estimates on the lower-N cells should be read accordingly. *Different theoretical orientation*: the charisma rubric grounds in Weberian charismatic authority (Weber, 1978) rather than Hegelian recognition. The two are correlated but not redundant; the §6.7 results do not displace recognition theory but document an architectural variant in which a complementary frame is also load-bearing. The §6.7 claims are framed as architectural exploration rather than as additional mechanisms in the three-mechanism model. Where the id-director extension introduces methodological novelty (the prompt-budget non-monotonicity, the rubric-curriculum alignment ablation, the cross-judge magnitude-compression observation), the contribution is descriptive and supplementary to the paper’s main argument; full audit trail is in the five `docs/cell-100-*.md` documents and the dialogue-log JSON for each pilot run.

The desire-for-recognition follow-up in §6.7 is narrower still. It is an iterative design slice, not

a confirmatory matrix: one run per listed profile-scenario row in the cells 159–169 authority-refusal search; selected static-profile diagnostics in cells 170–179; then a local resistance-breakthrough router line in cells 180–195. The first result is sufficient for a paper-side design finding (“accountable-bid charisma is the first clean local pass under authority refusal”) but not sufficient for a general claim about charismatic tutoring. The second result shows why a static-cell ladder is the wrong target: transfer demand, authority refusal, plain-language request, simplification follow-up, and resistance after scaffolding are different engagement states. The third result shows that an engagement-register router can work locally under the Codex/Claude stack (cell 193: 25/26 positive local mechanism outcomes across the combined eligible evidence) but also supplies the boundary: GLM 5.2 generation does not reproduce the mechanism, and even provider/runtime reasoning controls that produce clean DB validation leave the local breakthrough outcome at 0/5 on the five-signal check. A completed six-arm role-isolation grid now identifies the most plausible boundary as dynamic-learner completion and target drift rather than tutor/id public-register production: the GLM tutor/id with Codex learner arm remains locally usable (8/10 positive outcomes), the GLM dynamic-learner arms lose post-turn completion or target stability, and both scripted tutor controls route the public register with fixed resistant turns. The separate question-flood gate promotes cell 192 only for that subtype (2/2 answer-first usable commitments, 0 reopened or residual floods), not for the broader five-signal mechanism or GLM stack. To upgrade the finding after that pivot, the router would need a frozen grid spanning authority-withheld, status-challenge, partial-uptake, ordinary conceptual, vulnerability/persona-shift, transfer, plain-language, and instruction-to-engagement-switch scenarios; ≥ 10 repeats per profile-scenario as a pilot floor and preferably ≥ 24 for stable variance estimates; frozen comparators including cell 163, cell 104/107-style id-director baselines, the default tutor, and a no-router id-director baseline; at least two judges for both v2.2 and charisma scoring; model-pair variation across Codex/Claude, non-CLI hosted models, and smaller ego models; and a human-coded or human-learner subset focused on whether learners experience the tutor’s authority as defeasible rather than merely polished. Without those extensions, cells 169, 193, and the cell-192 question-flood subtype should be cited as exploratory design evidence motivating engagement-register routing, not as a generalizable effect.

The negative-register extension adds a further scope condition: arm assignment is not treatment fidelity. A cell can be configured as `sarcastic_challenge` while the generated tutor turn remains warm, ironic, or only superficially edgy. The hand-authored corrosive-exemplar check in §6.7 shows that the scoring instruments can detect corrosive sarcasm when fixed examples actually contain it, so the immediate limitation is not global judge blindness but generated-register fidelity. A fresh stance-gated smoke confirmed the practical scale of that limitation before repair: only 5/15 assigned negative-register rows counted as faithful arm evidence, while 10/15 were excluded as weak or warm-in-costume noncompliance and none were invalid person-attack violations. Later cue-repair canaries then showed that explicit stance-fidelity cues can restore treatment fidelity on sampled targets: 9/9 faithful, 0 invalid across boredom/irrelevance/rote-parroting, followed by 6/6 faithful, 0 invalid on held-out frustration/question-flood. Across the two post-repair canaries, all five resistance targets have now been sampled without gate exclusions or invalid person-attack violations; this is still an implementation check rather than an effect estimate. Negative-register claims

must therefore report both assigned-arm and faithful-arm estimands, and rows should contribute to a negative-register effect estimate only after the adopting tutor turn is classified as faithfully instantiating the assigned register contract. A related scope condition concerns the judge rather than the tutor: the register rubric’s own scores were originally produced by `openrouter.gpt-mini`, later shown to be numerically non-discriminating on this rubric (§6.7); every register-rubric conclusion for cells 196–198 now rests on a re-score under a sonnet-class judge (`claude-code.sonnet-5`), and the archived gpt-mini register-rubric figures are void and must not be cited. Sonnet-class judging is the standing requirement for the register rubric from the first row of any future negative-register run. Cells 196–198 remain a measurement-development addendum, not evidence about the pedagogical value or safety of sarcasm.

Neighbouring modules: Module 5 — Can Productive Tension Improve Teaching?; Module 1 — Meeting the Learner Raises the Floor.

Fazeli, S. et al. (2023). *Visual-verbal video analysis (VVVA): A multilayered qualitative framework for short-form video*. Method note.

Weber, M. (1978). *Economy and society: An outline of interpretive sociology* (G. Roth & C. Wittich, Eds.). University of California Press.